

Some Popular Models

Features and specifications	Hakko FX-888D	Okinternational MFR-1120	Weller WX 1	Goot Japan RX-802AS	MetroQ MTQ 937
					
Analogue/digital Wattage	Digital	NA	Digital	Digital	Digital
Overall	70W	70W	200W	80W	
Soldering pencil	65W (26V)	Output 60W Max at 22°C	40W	70W	48W
Temperature range	200 - 480°C	fixed temperature system that varies power to match the thermal demand of the device being soldered	50° to 550°C	50-450 degree	150°C to 450°C
Temperature stability	±1°C at idle temperature	±1.1°C in still air	±2°C	3°C	
Tip to ground potential	<2 mV	< 2 mV true RMS 50 to 500 Hz;			
Tip to ground resistance	<2 ohms	< 2 Ohms DC Unit On			
Thermal recovery	20 sec to reach 350°C			6 sec to reach set temp.	
Input/output voltage	220/26V AC	102 to 240V Input	230V, 50 Hz Input	220-240V input	220-240V/50 Hz
Heating element	Ceramic heater	Proprietary Skin Effect Heating/ Curie point control		Nichrome High Grade	
Iron stand	Yes	Yes (WS2)	Yes	Yes	
Dimensions	100×120×120mm	91×142×197mm	170×151×130mm	146×115×98mm	210×125×135mm
Cord length	1.2m	1.2m Burnproof ESD safe		1. 2m	
Standard tip	Shape-B (No.T18-B)	SXP Series cartridge Tips	RT3 (0054460399)	Pencil type (user can choose)	
Cleaning wire	Yes	Yes (Bronze Sponge)	Yes		
Cleaning sponge	Yes	Yellow wet sponge provided with the stand		Yes	Yes

more power and you would not like to change your iron every time.

Input/output voltage. Carefully check specifications of the soldering iron that you want to buy. Most irons run from the mains at 240V. However, there are soldering irons available that run with 120V supply too, which will not be suitable for India. If you are buying the soldering station from an authorised distributor, the one with 120V will obviously not be available in India. But some people make this mistake when they buy online from websites. Therefore do check the input/output voltage range.

Temperature accuracy. Temperature accuracy means how closely tip

temperature comes to the temperature that you have set at the station. There are several reasons due to which the temperature at the tip and the one set by you always differ. Sometimes designers compromise on accuracy to make the equipment cost-effective. Also, the tolerance for each component affects the accuracy to a great extent.

The size and shape of the tip also affect the accuracy, which can be explained by laws of thermodynamics and heat transfer. Tips lose heat by radiation, convection and conduction. The part that does the work is conduction and the other two are parasites. If the tip is thick with more surface area,

it can retain the heat better than long, thin ones.

The accuracy of ±5°C is good enough for any electronics work.

Temperature stability. This is a measure of how well the control circuit works, allowing heat loss to the environment.

Stability specification is valid only when the iron is at rest, not when it is actually used for soldering.

Thermal recovery. This too is a measure of how well the control circuit works. When a joint is soldered, heat is lost from the tip through conduction and the tip's temperature drops. Thermal recovery means, how long a soldering station will take with